

Multimode damping optimization of a long-span suspension bridge with damped outriggers for suppressing vortex-induced vibrations

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ABSTRACT

Long-span suspension bridges are subjected to wind-induced vibrations due to their large flexibility and low damping. Damped outriggers have been proposed recently to supplement damping and suppress multimode vibrations of main girders of long-span bridges, e.g., vortex-induced vibrations. However, when installed at multiple positions including tower-girder junction points and girder ends to further improve multimode damping effect, damping effects of the damped outriggers are interdependent. Additionally, the frequency curves of adjacent modes with respect to varying parameters of the damping devices could approach and intersect with each other, making it difficult to track mode orders and modal damping variations in design optimization. This study therefore proposes an optimization method to design damped outrigger parameters for maximizing multimode damping of the bridge. First, the bridge with damped outriggers is modeled generally using finite element method for dynamic analysis. Subsequently, two critical issues in the design, i.e., the mode tracking and multi-objective optimization, are respectively addressed by using the modal assurance criterion and the genetic algorithm. The proposed method is then applied to the Xihoumen Bridge with a main span of 1650 m. The results show that through the proposed optimization, 6 out of 7 modes vulnerable to vortex-induced vibrations can reach a damping ratio over 1.0%, which cannot be achieved by using damped outriggers of a uniform size. Furthermore, the required damper coefficients are decreased by 49.4%. The flexibility of the outriggers has been considered in the design. The developed method is promising in the design of other types bridge damping devices, e.g., multiple tuned mass dampers.

1. Introduction

Great progress has been made in the construction of long-span bridges, including cable-stayed bridges and suspension bridges in the last decades. With the increase in bridge span, bridge flexibility increases while bridge vibration frequency and damping decrease [1], making the bridge more vulnerable to vibrations induced by wind [2], earthquakes [3], vehicles [4] and so on. In recent years, several long-span bridges experienced vortex-induced vibrations (VIVs), such as the Xihoumen Bridge [5] and Humen Bridge [6] in China and Yi Sun-Shin Bridge [7] in Korea. Excessive vibrations could threaten the comfort of bridge users, and even endanger the safety of the bridge structure. Vibration problems of the bridges are regarded as a challenge which restricts the further development of bridge span [8]. Hence, bridge vibration mitigation is of great significance.

Existing vibration control methods for long-span bridges mainly include aerodynamic countermeasures, mechanical devices, and structural measures [9]. Among those, structural measures for vibration reduction entail the increment of structural mass and stiffness, which is not efficient for long-span bridges [10]. Aerodynamic countermeasures mitigate bridge vibrations by modifying cross-section shapes of the bridge girder, e.g., adding auxiliary components [11] to reduce aerodynamic forces on the bridge [12], or directly controlling the surrounding flow field, e.g., suction and jet control [13]. It can be realized by passive and active means. Nevertheless, the aerodynamic measures rely on wind tunnel tests to determine their effects, lacking a universal design method. With the increase of bridge span, aerodynamic countermeasures alone seem to be inadequate. Furthermore, the effects of aerodynamic measures are sensitive to structural damping. For example, the VIVs of the Humen Bridge were attributed to the damping decrease after the water-filled barriers were removed [14,15].

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In this context, vibration control of long-span bridges using mechanical devices becomes increasingly important.

Common mechanical countermeasures for vibration suppression of long-span bridges include mechanical dampers and tuned mass dampers (TMD) [16]. The TMD can be connected to a structure at an arbitrary location and absorbs vibration energy according to absolute displacement at the connection point of the structure [17]. Therefore, it has gained popularity in vibration control of long-span structures [18]. However, there are two critical issues of TMDs for long-span bridge vibration control. First, the TMD mass increases the load on the bridge girder. Secondly, a conventional TMD is often tuned to a particular mode, leaving the other modes under suboptimal control and still vulnerable to VIVs. To reduce the TMD mass, inerter is introduced to form a TMD inerter (TMDI) system for bridge vibration control [19,20]. The inerter can convert translation motion to high-speed rotation and hence amplify the inertial forces [21] such that the required physical mass is reduced. Meanwhile, the use of inerter decreases the static deformation of the spring, which can avoid large strokes of TMDs and satisfy the installation space limit in box girders. In addition, distributed tuned mass damper (DTMD) [22] and multiple tuned mass damper (MTMD) [23,24] have been proposed for robust control of single-mode and multimode vibrations. However, DTMD and MTMD further increase the load on the main girders. Besides, when more TMDs are installed, the stiffness and damping coefficients of each TMD need to be optimized according to the installation location, the added mass, and the target mode, leading to a complicated multi-parameter optimization problem.

Apart from TMDs, mechanical dampers have also been proposed to directly dissipate vibration energy of bridges. Such dampers have to be installed between two positions with relatively large displacements of the bridge to achieve sufficient damping. Otherwise, a damper support is required. For a suspension bridge with a main span and side spans, when the girder is not supported on the tower, vertical dampers can be directly installed between the tower and the girder [25,26]. Besides, the method of linking bridge towers, main girders and main cables of a suspension bridge with auxiliary cables is considered and the active tendons or dampers can be connected to the auxiliary cables in series for vibration mitigation [27,28].

Vibration control design for long-span bridges needs to take into account multiple modes as they can be excited under varied wind conditions [29,30]. Meantime, the design parameters include the installation locations and the device parameters. In other words, a multi-objective and multi-parameter optimization problem needs to be solved to achieve an optimal control. The TMD design for structural vibration control has been an active research topic [31,32]. For a single TMD added on the girder, the bridge may be simplified to a single-degree-of-freedom (SDOF) based on modal expansion, and then the well-established design formulas can be used [33]. For the DTMD and MTMD on a bridge, the genetic algorithm (GA) is mainly adopted for performance optimization, e.g., in wind resistant [22,34,35] and seismic design [23]. Other optimization algorithms have also been used, e.g., the hybrid genetic-simulated annealing algorithm [36], a combination of the GA and gradient algorithm for VIV control [37], the whale optimization algorithm for mitigating bridge-vehicle interactions [38], and firefly algorithm [39].

Recently, damped outriggers were proposed for bridge vibration mitigation [40,41], where the concept was verified by analytical and numerical analyses. This study further provides a method for optimal design of a suspension bridge with damped outriggers at multiple locations. The rest of this paper is divided into four sections. Section 2 introduces the bridge damped outriggers and the optimal design problem. Section 3 then presents the method for solving the optimization problem. In Section 4 the Xihoumen Bridge is taken as an example to demonstrate the proposed method, where the effect of outrigger stiffness on multimode damping ratios is also discussed. Concluding remarks are provided in Section 5.

2. Optimization problem of long-span bridges with damped outriggers

2.1. Damped outriggers for long-span bridges

Damped outriggers for long-span bridges have been proposed recently to supplement rotational damping to the main girders for energy dissipation and vibration mitigation [40]. Fig. 1 demonstrates a suspension bridge with damped outriggers consisting of outriggers and horizontal dampers placed in the longitudinal direction of the bridge. The outriggers are connected to the bridge girders near bridge towers/piers to convert girder rotations to translational displacements of their lower ends. The dampers dissipate vibration energy when there are relative motions between the outrigger ends and the bridge towers/piers during vertical vibrations of the girder. Multimode damping effects of the damped outriggers on long-span bridges have been investigated through theoretical and numerical analyses [40,42]. However, when damped outriggers are installed at multiple locations, the damper parameters need to be optimized with the consideration of their interdependence. The present study thus aims at optimization design of a suspension bridge with damped outriggers and the proposed method should apply to the design of bridges with other types of control devices, e.g., multiple tuned mass dampers.

2.2. Optimization problem

For vibration control of a suspension bridge with damped outriggers, the design parameters include the damper coefficients, outrigger stiffness, outrigger locations and boundary conditions, i.e., girder support conditions. They are discussed respectively in the following.

2.2.1. Damper coefficients

Linear viscous dampers (VDs) are considered for the damped outriggers. The viscous coefficients for each damper need to be determined. Considering a single damped outrigger installed on the bridge, there is a peak damping ratio for a target mode achieved by the optimal viscous coefficient [40]. When multiple damped outriggers are installed on the bridge for multimode vibration mitigation, it becomes a complicated multi-objective and multi-parameters problem to be addressed. An optimization algorithm is then applied herein to determine the optimal coefficients of the dampers at each position simultaneously.

2.2.2. Outrigger stiffness

Outrigger stiffness is also a key factor for the design of damped outriggers, which has been briefly analyzed using a simplified analytical model [40,42]. The results show that when the rigidity of outrigger attains a certain value, the maximum damping ratios and corresponding optimal damping coefficients of each mode would not increase with further increasing outrigger stiffness. It was also observed in modal analysis with the finite element (FE) model that the damping ratios of modes involving both longitudinal and vertical motions seem sensitive to outrigger stiffness. Therefore, the influence of outrigger stiffness is further studied using the FE model in design optimization.

2.2.3. Outrigger locations

Damped outriggers can be installed at both ends and intermediate positions of main girders close to the bridge towers/piers. They can achieve better damping effects when installed at the positions with large rotations during vibrations of the target modes. Generally, with the increased number of damped outriggers, multimode damping effects could be improved. In the optimization procedure, all feasible locations can be initially installed with damped outriggers, while those with viscous coefficients close to zero can be removed after optimization in the final design.



Fig. 1. The concept of damped outriggers installed on a long-span bridge.

2.2.4. Boundary conditions

For long-span bridges, boundary conditions can largely affect their dynamic characteristics. The bearings between the main girders and the towers/piers are of concern, particularly the vertical bearings. The influence of support conditions at both ends and intermediate positions of the main girder have been analyzed through a simplified analytical model [42]. In bridge design, the choice of support conditions can take into account the damping effects provided by control devices. However, for an existing bridge, support conditions between the girders and towers/piers are given and normally difficult to change.

3. Design optimization method

3.1. Modeling of long-span bridges with damped outriggers

The FE method is widely used to discretize a continuous system with infinite degrees of freedom into a system with finite degrees of freedom for structural analysis [43]. Compared with the simplified analytical model [40], the FE model is generally applicable and able to consider detailed design parameters, e.g., boundary constraints, lateral supports between main girders and towers, the interaction between main cables and the main girders, and so on.

Without loss of generality, the linear dynamic motion of the system after discretization is governed by the equation

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f}, \quad (1)$$

where $\mathbf{M}$, $\mathbf{K}$ represent the mass and stiffness matrices of the structure, respectively, $\mathbf{C}$ is the structural damping matrix, $\mathbf{u}$ denotes the nodal displacement vector, and $\mathbf{f}$ is the external load vector. Correspondingly, $\dot{\mathbf{u}}$ and $\ddot{\mathbf{u}}$ denote the velocity and acceleration vectors, respectively. The inherent damping is ignored in this study. Therefore, when the damping coefficients of damped outriggers are zero, the damping matrix $\mathbf{C} = \mathbf{0}$ with $\mathbf{0}$ denoting a matrix of zeros. The frequencies and mode shapes of the bridge are real numbers. To consider the geometric nonlinear effect caused by the deformation of the main cables under the dead load, static analysis of the bridge is performed before modal analysis.

In this study, damping ratios of target modes are focused to appreciate the vibration control effects of long-span suspension bridges. According to previous wind tunnel tests [44], the amplitudes of bridge under VIVs can meet the design requirement when the damping ratios of target modes are above 1.0%. As aforementioned, structural inherent damping is ignored and dampers of the damped outriggers are the only source of damping in the FE Model when damped outriggers are installed. In this situation, the structural damping matrix $\mathbf{C}$ has only nonzero entries associated with the damper elements. Eigen-analysis of Eq. (1) is carried out to obtain modal properties including frequency, damping and mode shapes.

The bridge with damped outriggers is a typical system with non-classical damping, leading to a complex eigenvalue analysis problem. Hence, the damped method which uses the full matrices ($\mathbf{M}$, $\mathbf{K}$, and $\mathbf{C}$) to calculate the complex eigenvalues and corresponding eigenvectors directly with Lanczos Algorithm [45] is adopted in FE analysis. Eventually, the computed complex eigenvalues can be expressed as

$$\lambda_n = \sigma_n \pm i\omega_n, \quad (2)$$

in which λ_n represents the n th order complex eigenvalues, $i = \sqrt{-1}$ is imaginary unit, σ_n and ω_n are the real and imaginary parts of the eigenvalues, respectively. Furthermore, ω_n also represents the n th order damped frequency and the corresponding damping ratio ζ_n can be calculated as

$$\zeta_n = \frac{-\sigma_n}{|\lambda_n|} = \frac{-\sigma_n}{\sqrt{\sigma_n^2 + \omega_n^2}}. \quad (3)$$

The eigenvectors correspond to the mode shapes of the bridge vibrations.

3.2. Mode tracking

It is important to compare damping of the identical modes with varying parameters, e.g., damper coefficients, outrigger stiffness, and so on. The adjacent modal frequency curves versus a changing design variable could approach and intersect with each other, posing difficulty in ordering the modes. For parameter optimization, this mode tracking issue is a critical problem to be addressed. Herein, the modal assurance criterion (MAC) is adopted for tracking mode orders by comparing mode shapes with varying parameters of the damping devices. The criterion has been used to assess the correlation between two mode shapes, e.g., in the field of modal identification to distinguish real modes from fictitious modes [46,47].

Let the mode shapes of a bridge with damping devices extracted by eigen-analysis be denoted by $\{\phi_i\}$ with i the mode order. For optimization, one or more parameters are varied and the modal analysis is repeated leading to a new set of mode shapes, denoted by $\{\psi_j\}$ with j the mode order in this case. The correlation of the two modes is appreciated by using MAC value computed as [48]

$$\text{MAC}(\{\phi_i\}, \{\psi_j\}) = \frac{|\{\phi_i\}^T \{\psi_j\}|^2}{\{\phi_i\}^T \{\phi_i\} \{\psi_j\}^T \{\psi_j\}}, \quad (4)$$

where $\{\}^T$ represents the transpose of original vector. Note that MAC value is in the range of $[0, 1]$. When it is zero, there is no correlation between the two modes; when MAC value is 1, the corresponding two modes are identical. For a particular structure, mode shapes corresponding to different modes are orthogonal, which means MAC values between two different modes are always zero. For a structure with

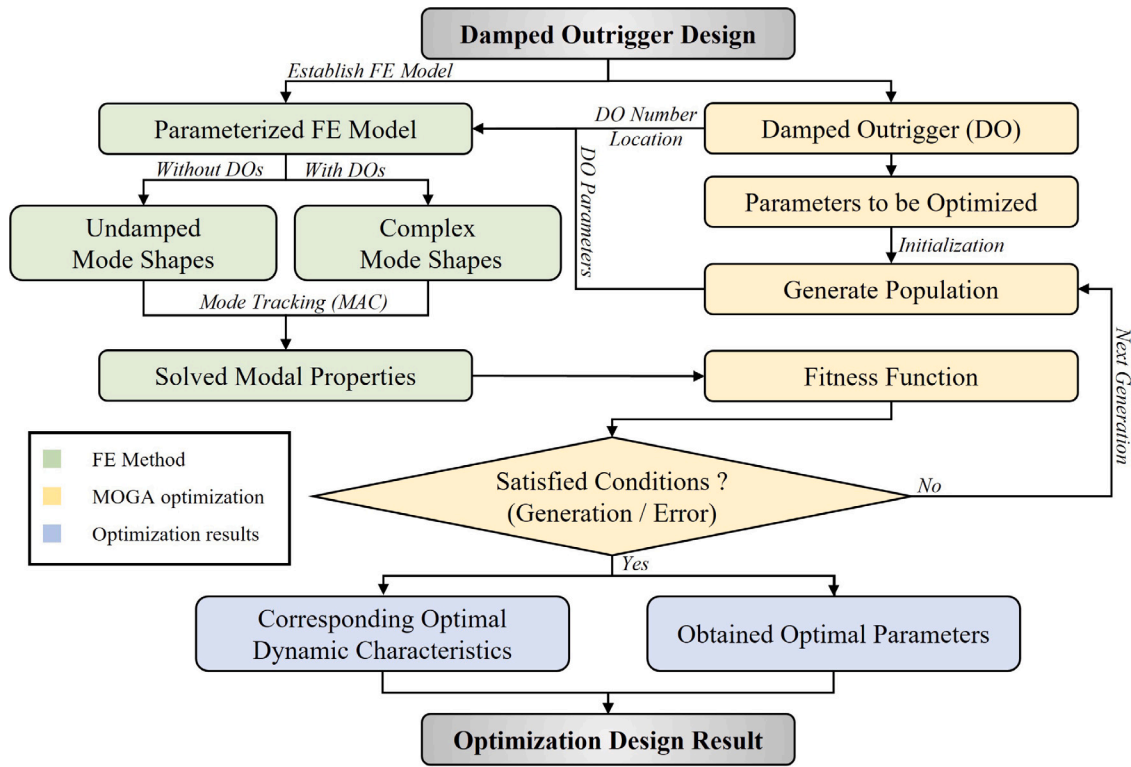


Fig. 2. Flowchart of damped outrigger optimization process for vibration mitigation.

design parameters varying in certain ranges, the change of mode shapes can be assumed relatively small. Hence, MAC value between the altered mode shape vector and the corresponding original mode shape vector approaches 1. The corresponding vibration modes of the structure with varying parameters are thus tracked.

Furthermore, with damped outriggers installed on the bridge, the bridges have complex-valued frequencies and mode shapes due to the non-proportional damping. MAC values between two complex mode shapes are computed by [49]:

$$MAC(\{\phi_i\}, \{\psi_j\}) = \frac{|\{\phi_i\}^T \{\psi_j^*\}|^2}{\{\phi_i\}^T \{\phi_i^*\} \{\psi_j\}^T \{\psi_j^*\}}, \quad (5)$$

where “*” represents the conjugate of the original vector.

For a bridge structure with negligible inherent damping, the corresponding modes are tracked by comparing the real mode shapes without damping (damped outrigger with zero damper coefficient) and the complex mode shapes when damper coefficients are nonzero. Hence, Eq. (5) is adopted for computation by regarding the real mode shapes as complex mode shapes with zero imaginary parts.

3.3. Genetic algorithm for solving the optimization problem

The previously mentioned multi-objective and multi-parameter problem can be solved by typical optimization algorithms [50,51] including GA, particle swarm optimization, simulated annealing algorithm, and so on. Besides, to ensure that both the total damping ratio and the minimal damping ratio of all concerned modes are sufficiently large, more than one objective functions are required. In this case, the multi-objective genetic algorithm (MOGA) [52] is desired. MOGA is able to find a set of optimal Pareto Fronts within the objective range of the solution so that it can meet the multiple objective functions and achieve a better level, as showed in the design optimization of cross-ties and dampers for vibration mitigation of stay cables [53,54]. Hence, MOGA is applied in this study. Furthermore, the ranges of design parameters need to be prescribed. For the dampers, the viscous

coefficients can vary between zero to a relatively large value. Notably, for seismic response mitigation, viscous dampers with high viscous coefficients have been used for bridges, e.g., up to 3×10^4 kN s/m [55]. For the outrigger stiffness, a critical value can be determined using the simplified analytical model [40] over which the damping effect increases slowly. Then, the stiffness is allowed to vary in a range around this value to reduce the computation cost in optimization.

3.4. Automated optimization procedure

The design and parameter optimization steps of damped outriggers in practical engineering are described as follows:

- (1) A parameterized finite element model of the bridge with damped outriggers is built.
- (2) Parameters of the damping system are initialized to zero to calculate the real mode frequencies and shapes of the bridge. The modal properties of target modes are extracted and recorded for mode tracking.
- (3) MOGA is then applied for parameter optimization.

Fig. 2 shows the flowchart of the optimization process.

Among the design parameters discussed in Section 3.3, the damper coefficients are optimized by MOGA in this study. The influences of outrigger stiffness and locations are also discussed but they are not directly optimized by MOGA. First, outriggers with small stiffness lead to bridge modes featured by large longitudinal movements that need to be avoided in design. Secondly, when the outrigger stiffness is larger than a certain value, further increase of stiffness has small influence on damping, as will be shown in the subsequent section. Different combinations of practical outrigger locations are investigated respectively.

4. Case study

4.1. The Xihoumen Bridge

The Xihoumen Bridge (Fig. 1) is a suspension bridge [56] connecting the Cezi Island and Jintang Island in Zhoushan, China. The bridge

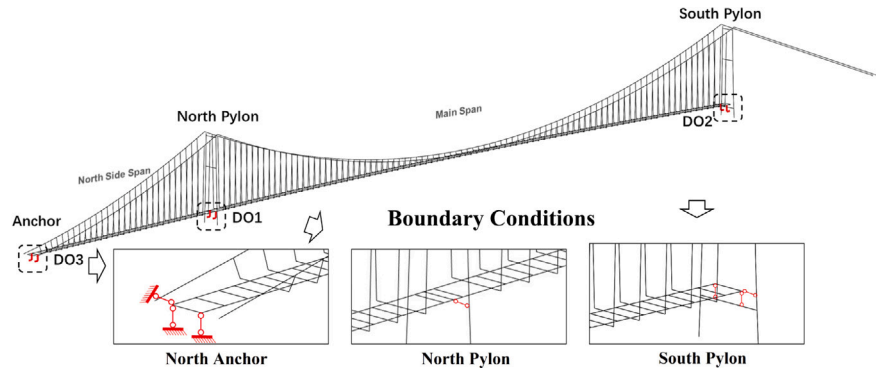


Fig. 3. Finite element model of the Xihoumen Bridge with damped outriggers.

is located at a typhoon-prone area. Therefore, a slotted twin-box main girder was designed for wind resistant. The bridge has a main span of 1650 m and a north side span of 578 m. Vertical constraints of the bridge girder are provided at the north anchor and the south pylon, and there are no vertical bearings between the main girder and north pylon. Horizontal dampers are installed respectively at the north anchorage and the south pylon to limit the longitudinal movement of the girder. In this study, since the installation of damped outriggers at both ends can serve the function of the original longitudinal dampers which are thus not included in the FE model.

The selection of the twin-box steel main girder brings critical wind speed of the flutter to 90 m/s, which further improves the aerodynamic stability of the bridge [57]. Nevertheless, compared with the single-box girder, vortex excitation problem in this case is more complicated and the bridge is more prone to VIVs [58]. Multiple VIV events occurred to the bridge according to field monitoring [5,29,59]. Meanwhile, the damping ratios of the modes subjected to VIVs are found small (less than 0.5%). Therefore, it is of great practical significance to improve multimode damping of the Xihoumen Bridge.

4.2. Finite element model

Fig. 3 illustrates the FE model of the Xihoumen Bridge. The whole bridge model consists of 1882 nodes and 2900 elements. ANSYS Software [45] is adopted for establishing the model and analyzing the modal properties. BEAM4 elements with user-defined area and area moment of inertia are used for the main girders and pylons. Since the main girder of the Xihoumen Bridge is a slotted twin-box girder, two parallel beams are built and they are connected by cross beam elements. Meanwhile, LINK10 elements with user-defined area and initial strain are adopted for the suspenders and main cables of the bridge. The initial strain is calculated from the design cable force given in the structural drawings. The main cables are connected to two pylons at the top and their north ends are fixed at the north anchor.

There are three possible positions for the installation of damped outriggers, i.e., the north pylon (DO1), the south pylon (DO2), and the north anchor point (DO3), as shown in Fig. 3. To fit the original design of the bridge, the damped outriggers are connected to the girder at the position of the cross-beam closest to the tower/anchor. At each installation position, two damped outriggers are respectively mounted on the two main beams of the FE model. The length of each vertical outrigger is 15 m. Furthermore, the damping coefficients and outrigger stiffness of three damped outriggers can be expressed as c_{oj} ($j = 1, 2, 3$) and EI_{oj} ($j = 1, 2, 3$), respectively.

Fig. 4 shows mode shapes of the first 14 modes of the bridge without damping (all dampers have zero coefficients). As the performance of damped outriggers mainly depend on the rotation amplitudes of the main girders, the main girder rotations of each mode are represented by red dotted lines in Fig. 4 and the circles indicate the damped outrigger locations. Accordingly, the observed VIV modes correspond to the 5th,

6th, 8th, 9th, 10th, 12th and 13th vertical modes from FE analysis. These 7 modes are marked in the figure. In addition, the 1st mode entails both vertical and longitudinal vibrations with a frequency of 0.1106 Hz. Nevertheless, the longitudinal vibration component vanishes when there is sufficient damping provided by damped outriggers, and the frequency is decreased to 0.0945 Hz.

4.3. Mode tracking performance

The modes are ordered according to frequency (or the magnitude of complex frequency) from FE analysis. It is noted that modal frequency of this bridge are closely-spaced and mode order interchange can occur between adjacent modes. The performance of MAC for mode tracking is first evaluated.

The modes of the bridge with zero damping are chosen as reference modes with mode shapes denoted by $\{\phi_{ref}\}$. The modes of the bridge with varied parameters in damped outriggers are considered to be reordered and the corresponding mode shapes are denoted by $\{\phi_{obj}\}$ which is complex-valued. Therefore, MAC values are then calculated using Eq. (5). When considering the damped outrigger located only at DO2 with damping coefficient of $c_{o2} = 0.32 \times 10^5$ kN s/m (optimal damping coefficients as depicted in Ref. [40]) and rigid outrigger, MAC values of the complex mode shapes and the undamped mode shapes of the first 14 modes are showed in Fig. 5 (a). It can be seen that MAC values corresponding to the same vertical bending modes approach 1, and MAC values between the corresponding objective mode and other modes are almost zero. Interestingly, as shown in Fig. 5 (a), the 1st and 2nd modes (reference mode shape $\{\phi_{ref}\}$) are both coupling modes of longitudinal movement and vertical bending, and the longitudinal movement vanishes with a certain amount of damping. Thus, MAC values between these two modes are slightly higher than the other modes, which nevertheless does not affect the accuracy of mode tracking. Furthermore, Fig. 5 (b) plots the evolution of selected MAC values along with increasing viscous coefficient of DO2, suggesting an excellent performance in tracking the 6th mode.

4.4. Optimization of single damped outrigger

4.4.1. Damping coefficient of damped outrigger

The cases of adding damped outriggers at three positions respectively have been investigated [40] and Table 1 lists the results. The viscous coefficients of DO1, DO2 and DO3 which achieve the optimal damping ratios are $c_{o1} = 1.04 \times 10^5$ kN s/m, $c_{o2} = 0.32 \times 10^5$ kN s/m and $c_{o3} = 0.162 \times 10^5$ kN s/m, respectively. The outriggers are assumed rigid.

As shown in Table 1, 6 out of 7 VIV modes (6th, 8th, 9th, 10th, 12th, and 13th) can reach the damping ratios over 0.98% with the damped outrigger installed at DO2, which roughly meet the requirement of 1.0%. Since the damped outrigger provides small damping to the 5th mode regardless of damper parameters because of small rotations at all

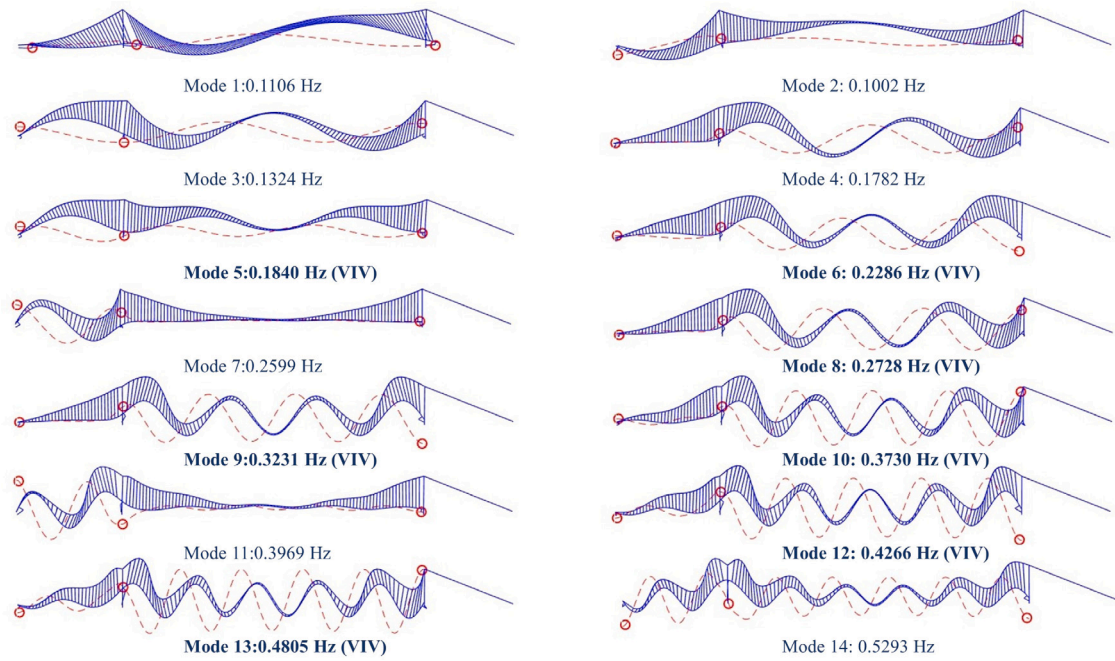


Fig. 4. The first 14 vertical bending modes and corresponding rotations of the Xihoumen Bridge.

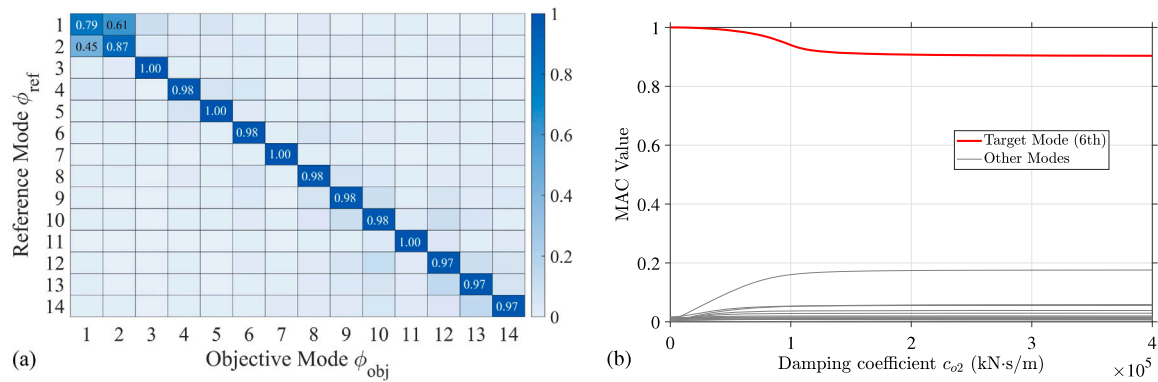


Fig. 5. MAC values: (a) between objective modes and reference modes when rigid DO2 with damper coefficient $c_{o2} = 0.32 \times 10^5$ kN s/m is installed; (b) between objective modes and the 6th reference mode when rigid DO2 with varying viscous coefficient is installed.

Table 1

Damping effects of the Xihoumen Bridge with one rigid outrigger as compared to the measured damping ratios of the bridge without damped outriggers.

Vertical mode no.	Observed VIV Events	Measured (without DOs) [5,29]		Computed results [40]				
		Frequency		Frequency (Hz)		Damping ratio (%)		
		(Hz)	ratio (%)	Undamped	Damped (DO1)	DO1	DO2	DO3
1		0.095	0.57	0.1106	0.0945	1.21	3.65	9.90
2		—	—	0.1002	0.1009	0.11	0.61	1.21
3		0.133	0.52	0.1324	0.1315	0.10	0.01	0.00
4		—	—	0.1782	0.1778	0.11	0.61	0.00
5	Yes	0.183	0.50	0.1840	0.1838	0.06	0.06	0.14
6	Yes	0.230	0.51	0.2286	0.2277	0.58	1.13	0.01
7		—	—	0.2599	0.2570	2.62	0.00	1.47
8	Yes	0.276	0.39	0.2728	0.2718	0.51	1.08	0.01
9	Yes	0.324	0.42	0.3231	0.3249	1.01	1.27	0.00
10	Yes	0.378	—	0.3730	0.3746	0.24	1.17	0.01
11		—	—	0.3969	0.4062	1.35	0.02	2.22
12	Yes	0.435	0.37	0.4266	0.4306	0.81	1.12	0.03
13	Yes	0.493	0.32	0.4805	0.4825	0.21	0.98	0.05
14		—	—	0.5293	0.5302	0.05	0.44	0.52

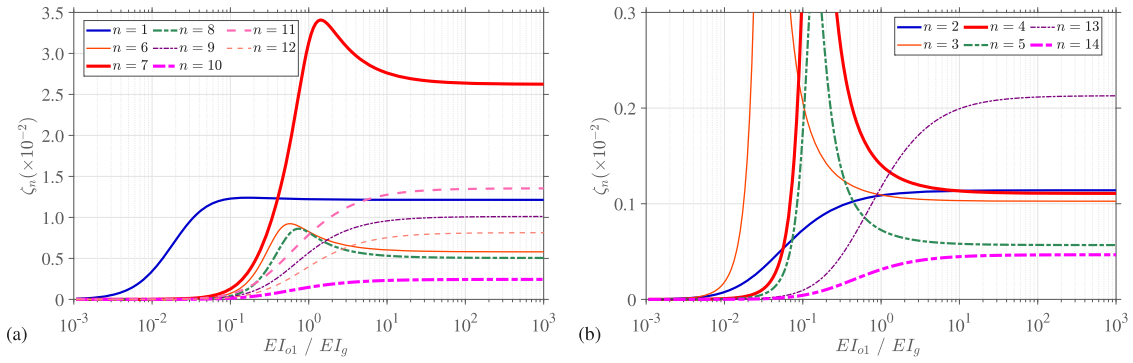


Fig. 6. Relationship between bridge damping ratios and outrigger stiffness when DO1 is installed with the viscous coefficient $c_{o1} = 1.04 \times 10^5$ kN s/m.

possible installation locations (see Fig. 4), the other six modes vulnerable to VIVs are targeted in parameter optimization in the following investigation. Nevertheless, the damping effects of target modes could be reduced considering flexibility of the outrigger, as discussed in the following.

4.4.2. Influence of outrigger stiffness

The influence of outrigger bending stiffness has been discussed using a simplified analytical model [40]. In this section, the effects are further studied using the FE method.

Installation of DO1. Fig. 6 shows variations of modal damping ratios with respect to changing outrigger bending stiffness (EI_{o1}), when DO1 is installed and the viscous coefficient $c_{o1} = 1.04 \times 10^5$ kN s/m. The value of $c_{o1} = 1.04 \times 10^5$ kN s/m is chosen through maximizing the damping ratios of the VIV modes of the bridge with one rigid outrigger. A stiffness ratio is defined as EI_o/EI_g with EI_g the vertical bending stiffness of the girder, which ranges from 10^{-3} to 10^3 . It is notable that when the stiffness ratio equals 10^3 , the damping ratios of the modes are identical to the corresponding damping ratios in the case of the bridge with a rigid outrigger in Section 4.4.1, indicating that the stiffness ratio is sufficiently large.

To better illustrate the influence of outrigger stiffness, an efficiency factor, denoted by ρ , is defined as follows,

$$\rho = \frac{\zeta}{\zeta_r} \times 100\%, \quad (6)$$

where ζ_r indicates the damping ratio when the outrigger is rigid. When ρ approaches 100%, the outrigger stiffness is considered sufficient. A less rigid outrigger can lead to an apparent change in the mode shapes and hence significantly affect the attainable damping ratios. For example, the two modes shown in Fig. 7 are dominated by vertical vibrations when the outriggers are rigid, but are dominated by longitudinal vibrations when the outrigger stiffness decreases. As the dampers are installed in longitudinal direction, much larger damping is achieved for modes dominated by longitudinal vibrations, thus leading to ρ exceeding 100%. Notably, when the stiffness ratio (EI_o/EI_g) is 2, the efficiency factor ρ for the first 14 modes ranges from 70.3% to 127.6%, suggesting that it is necessary to take outrigger stiffness into consideration.

Installation of DO2. When DO2 is installed, Fig. 8 depicts the damping curves with varying bending stiffness ratio. The viscous coefficient, $c_{o2} = 0.32 \times 10^5$ kN s/m, is the optimal damper coefficient determined by considering rigid outriggers. In this case, when the damped outrigger stiffness decreases, longitudinal components in some modes increase but their amplitudes are not too large overall, and hence the peak damping is close to the damping in the case of rigid outrigger, as shown in Fig. 8. The damping variations are quite consistent with the results obtained by using a simplified model [40]. Furthermore, when the stiffness ratio (EI_{o2}/EI_g) is 2, the efficiency factor ρ for the first 14 modes changes from 88.1% to 104.2%, which is close to 100%, indicating sufficiently large outrigger stiffness.

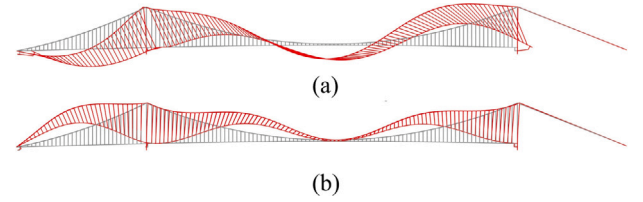


Fig. 7. Coupling modes of longitudinal movement and vertical bending of the bridge with DO1 installed: (a) Mode 3 when the stiffness ratio is 0.036; (b) Mode 5 when the stiffness ratio is 0.145.

Installation of DO3. Fig. 9 shows the influence of outrigger stiffness when DO3 is installed on the bridge and the optimal damping coefficient $c_{o3} = 0.162 \times 10^5$ kN s/m is determined assuming rigid outrigger. It is seen that DO3 provides relatively large damping to modes dominated by side-span vibrations while it has limited effects on the target modes which are mainly dominated by vibrations of the main span. The stiffness ratio is also determined as 2 and correspondingly ρ varies from 93.1% to 101.7% for the 14 modes.

Table 2 lists the damping ratios provided by the damped outriggers respectively installed at the three location with $EI_{oj}/EI_g = 2$ ($j = 1, 2, 3$), together with ρ . It is seen that most of the modes subjected to VIVs can attain a damping comparable to that in the limit of rigid outriggers. Therefore, the stiffness ratio of 2 is recommended for practical design.

4.5. Optimization of damped outriggers at two or more locations

In this case, the optimization design of multiple damped outriggers installed on the bridge is considered. The damper coefficients, outrigger stiffness and outrigger locations are mainly concerned. The damper coefficients are first optimized through MOGA considering rigid outriggers. The influence of the outrigger stiffness is discussed afterwards. The following two design methods in determination of the optimal damper coefficients are compared to emphasize the importance of the proposed optimization method:

- (1) Equal damper coefficients. Simply use dampers of uniform size at different positions, i.e. $c_{o1} = c_{o2} = c_{o3}$. Therefore, the multiple damper coefficients are transformed into an optimization problem with only one independent variable.
- (2) Damper coefficients optimized simultaneously through MOGA. MOGA is used to determine the optimal damper coefficients, as presented in Section 3.3. The objective functions are defined as

$$f(c_{o1}, c_{o2}, c_{o3}) = \sum_{n=1}^N \zeta_n \quad (N = 14), \quad (7)$$

$$g(c_{o1}, c_{o2}, c_{o3}) = \min(\zeta_6, \zeta_8, \zeta_9, \zeta_{10}, \zeta_{12}, \zeta_{13}),$$

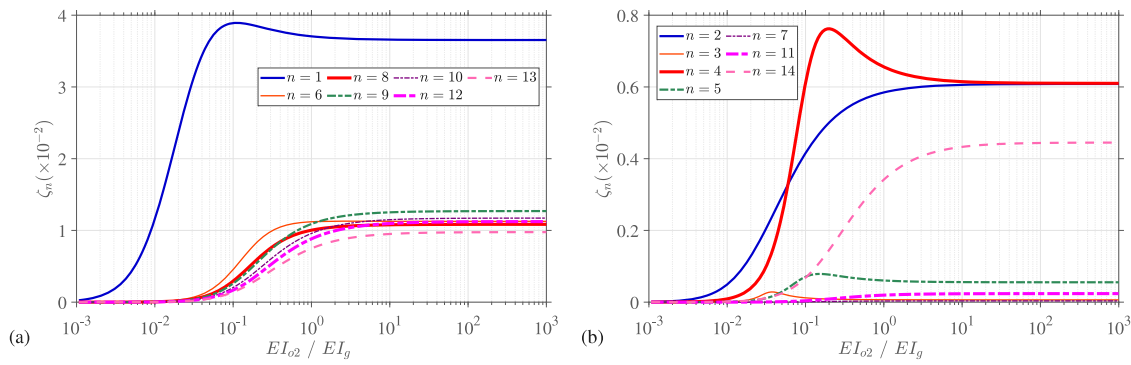


Fig. 8. Relationship between bridge damping ratios and outrigger stiffness when DO2 is installed with the viscous coefficient $c_{o2} = 0.32 \times 10^5$ kN s/m.

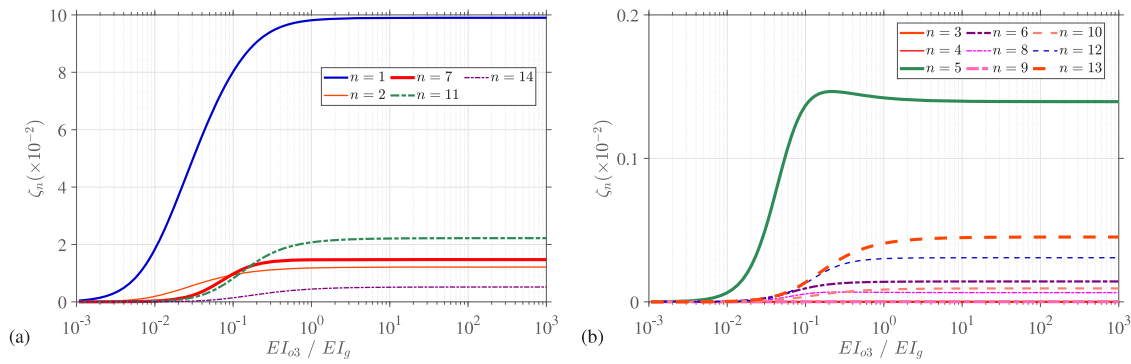


Fig. 9. Relationship between bridge damping ratios and outrigger stiffness when DO3 is installed with the viscous coefficient $c_{o3} = 0.162 \times 10^5$ kN s/m.

Table 2

Damping effect and efficiency factor of the Xihoumen Bridge with a single damped outrigger when the stiffness ratio is 2.

Vertical mode no.	DO1			DO2			DO3		
	EI_{o1}/EI_g	Efficiency factor (%)		EI_{o2}/EI_g	Efficiency factor (%)		EI_{o3}/EI_g	Efficiency factor (%)	
	2	∞		2	∞		2	∞	
1	1.22	1.21	100.3	3.68	3.65	100.7	9.87	9.90	99.7
2	0.11	0.11	97.8	0.60	0.61	98.2	1.20	1.21	98.8
3	0.11	0.10	102.9	0.01	0.01	102.8	0.00	0.00	101.7
4	0.12	0.11	111.5	0.63	0.61	103.6	0.00	0.00	100.9
5 ^a	0.06	0.06	111.8	0.06	0.06	104.2	0.14	0.14	100.9
6 ^a	0.69	0.58	119.7	1.13	1.13	100.2	0.01	0.01	99.2
7	3.26	2.62	124.1	0.00	0.00	96.3	1.47	1.47	99.9
8 ^a	0.65	0.51	127.6	1.05	1.08	96.9	0.01	0.01	101.5
9 ^a	0.77	1.01	76.6	1.19	1.27	93.4	0.00	0.00	98.5
10 ^a	0.19	0.24	78.0	1.07	1.17	91.2	0.01	0.01	96.0
11	1.04	1.35	76.6	0.02	0.02	91.4	2.16	2.22	97.1
12 ^a	0.57	0.81	70.3	1.00	1.12	89.3	0.03	0.03	99.3
13 ^a	0.16	0.21	74.3	0.86	0.98	88.2	0.04	0.05	95.5
14	0.04	0.05	82.2	0.39	0.44	88.1	0.48	0.52	93.1

^arepresents the modes subjected to VIVs according to the monitoring data in previous research [5,29].

where f is to maximize the total damping ratio of the first 14 vertical modes on the Xihoumen Bridge, and objective function g aims at improving the lowest damping of 6 out of 7 modes subjected to VIVs except the 5th mode. The damped outriggers cannot supply sufficient damping to the 5th mode regardless of the damper coefficients, owing to small rotations of the girder at all the three installation points (Fig. 4). Besides, the range of damping coefficients at each position vary from 0 to 4×10^5 kN s/m, which can ensure the optimal damping coefficient values are covered.

4.5.1. Damped outriggers at two positions

Since modes subjected to VIVs are mainly dominated by vibrations in main span, when considering two damped outriggers installed on the bridge, it is preferred to install DO1 and DO2. The damper coefficients

are optimized at first through the above two methods considering rigid outriggers. Subsequently, the influence of stiffness is investigated with the optimal damper coefficients obtained by Method 2.

When the coefficients of the two damped outriggers are assumed of the same, it was roughly determined that the optimal damper coefficients are $c_{o1} = c_{o2} = 0.36 \times 10^5$ kN s/m [40]. Correspondingly, the damping ratios of the first 14 modes are listed in Table 3, named as Case 1, for further comparison. MOGA optimization (labeled Case 2) gives the optimal damper coefficients that $c_{o1} = 0.066 \times 10^5$ kN s/m and $c_{o2} = 0.34 \times 10^5$ kN s/m. Table 3 lists the corresponding damping ratios.

With the damper coefficients same as Case 2, Fig. 10 plots the efficiency factor ρ influenced by the varying outrigger stiffness for the 6 target modes. Subsequently, the stiffness ratios of DO1 and DO2 are determined as 2, and the corresponding efficiency factors are marked in Fig. 10, respectively. Furthermore, the damping ratios of the first 14

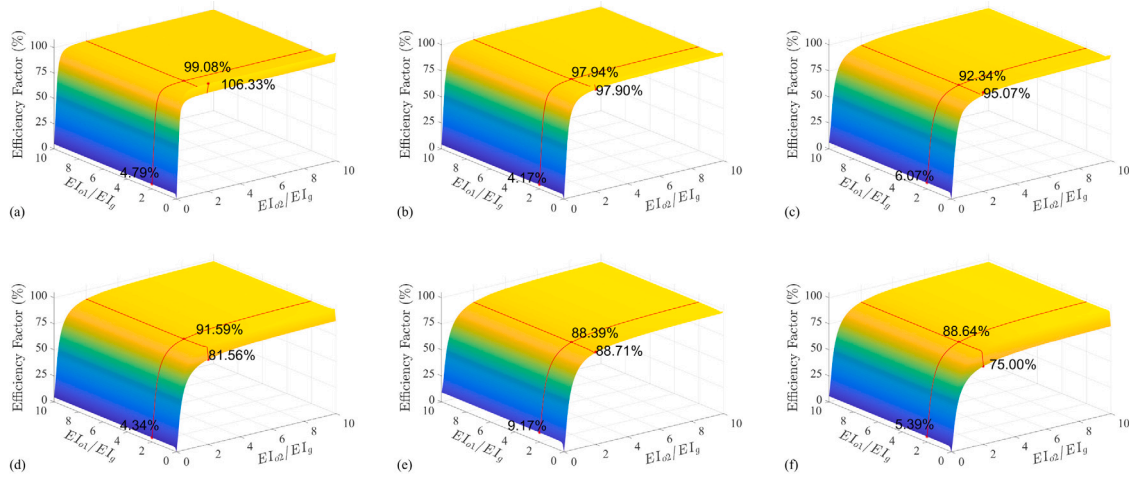


Fig. 10. Efficiency factor ρ of target modes with different stiffness ratio when DO1 and DO2 are installed: (a) 6th mode; (b) 8th mode; (c) 9th mode; (d) 10th mode; (e) 12th mode; (f) 13th mode.

Table 3
Optimal damping ratio of the Xihoumen Bridge with two damped outriggers.

Mode No.	Frequency (Hz)	Damping ratio (%)			Efficiency factor ρ (%)
		Case 1	Case 2	Case3	
1	0.0945	1.60	2.82	2.84	100.7
2	0.1009	0.37	0.50	0.50	98.3
3	0.1315	0.96	0.33	0.33	99.5
4	0.1778	0.31	0.53	0.55	103.6
5 ^a	0.1838	0.04	0.05	0.05	104.3
6 ^a	0.2277	1.03	1.13	1.12	99.1
7	0.2570	0.66	0.16	0.16	100.7
8 ^a	0.2718	0.96	1.13	1.11	97.9
9 ^a	0.3249	1.15	1.28	1.18	92.3
10 ^a	0.3746	2.04	1.32	1.21	91.6
11	0.4062	1.32	0.31	0.31	100.4
12 ^a	0.4306	1.03	1.12	0.99	88.4
13 ^a	0.4825	1.85	1.12	1.00	88.6
14	0.5302	1.26	0.61	0.55	90.2

^arepresents the modes subjected to VIVs.

modes are all listed in Table 3 as Case 3. To sum up, for the 6 target modes, with the optimized damper coefficients and practical outrigger stiffness, the target modes can achieve damping ratios over 1.0% to fulfill the design requirements.

Table 3 compares the achieved damping ratios in Case 1, Case 2, and Case 3. The results are further compared in Fig. 11, where the red dash line represents the required damping ratio, i.e., 1.0%. It can be found the optimization by MOGA (Method 2) has two advantages. First, it can significantly decrease the damper coefficients. The total damper coefficient decreases by 43.6%, making it more practically feasible, through the comparison of damper coefficients in Case 1 and Case 2 (Table 4). The optimization lead to several candidate groups of damper coefficients that can achieve comparable damping performances. The candidate group with a smaller total damper coefficient is selected manually for feasibility consideration. It is also found the damper coefficient of DO2 is much larger than that of DO1. Recall that the DO2 when installed individually performs well in multimode damping enhancement. Therefore, in the case of combined DO1 and DO2, the contribution of DO2 dominates and the optimal damping coefficient of DO2 is closer to that in the case of only DO2 installed on the bridge. Furthermore, Method 2 makes more modes achieve the damping ratios over 1.0%. Specifically, the modal damping ratios of the 6th, 8th and 12th modes are considerably increased by using MOGA optimization compared with Method 1. Note that the damping ratios of high-damping modes like the 10th and 13th modes are decreased

to some extent but still over 1.0%, meaning that balanced multimode damping effects are achieved with reduced cost (smaller damper size).

4.5.2. Damped outriggers at three positions

When installing damped outriggers at three positions, the damper coefficients are optimized through Method 1 and Method 2 with rigid outrigger, coined as Case 1 and Case 2. The damper coefficients in Case 3 are same as Case 2 but considering practical outrigger stiffness.

First, the damping curves of the first 14 modes with equal damper coefficients are depicted in Fig. 12. Optimal damper coefficients are determined as $c_{o1} = c_{o2} = c_{o3} = 0.56 \times 10^5$ kN s/m, and the corresponding damping ratios of the first 14 modes are listed in Table 5. With the same objective functions as in Section 4.5.1, MOGA optimization gives the optimal damper coefficients that $c_{o1} = 0.19 \times 10^5$ kN s/m, $c_{o2} = 0.64 \times 10^5$ kN s/m, and $c_{o3} = 0.196 \times 10^5$ kN s/m, respectively. The total damper coefficient through optimization decreases by 49.4% compared with that in Case 1 as listed in Table 4.

It is worth noting that total damper coefficient with three damped outriggers has a significant increase comparing with two damped outriggers installed on the bridge. A comparison of Figs. 11 and 14 (both Case 1 and Case 2) shows that three damped outriggers can provide better damping effects for the 5th, 7th 10th, 11th and 14th modes, which leads the optimal damping coefficients closer to the optimal damping coefficients for these modes. Since these modes require a larger damping coefficient, the total damping coefficient is also larger. The damping ratios with optimal damper coefficients and rigid outriggers are listed in Table 5.

The influence of outrigger stiffness is studied in the ranges of $EI_{o1}/EI_g \in [0, 10]$ and $EI_{o2}/EI_g \in [0, 10]$. While $EI_{o3}/EI_g = 2$ is chosen because DO3 is mainly effective for modes dominated by side span and such amount of stiffness is found sufficient in Section 4.4.2. Fig. 13 plots the efficiency factor ρ in the concerned ranges for the target modes. It is seen that the stiffness ratios can still be designed as 2. Furthermore, Table 5 lists ρ for all the 14 modes when $EI_{oj}/EI_g = 2$ ($j = 1, 2, 3$), and the comparison of damping effects in different cases is displayed in Fig. 14.

As listed in Table 5, damping improvements when adding DO3 can only found in the modes dominated by the side span, e.g., the 7th and 11th modes. Meanwhile, the minimal damping ratio of target modes is further increased to about 1.1% in Case 2 and Case 3. Notably, for Case 1, the minimal damping ratio of target modes is 0.81% (the 8th mode), which is even smaller than that achieved by two damped outriggers (Table 3). It is attributed to the interactions among the three damped outriggers. Therefore, the optimization through MOGA is of importance

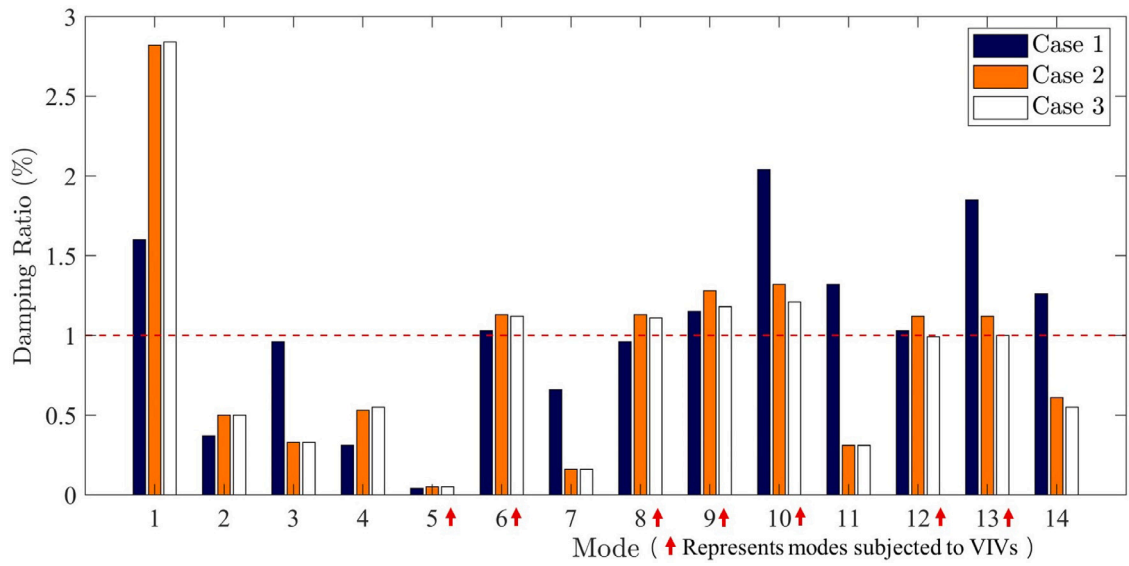


Fig. 11. A comparison of multimode damping effects realized by two damped outriggers with equal damper coefficients and with simultaneously optimized damper coefficients.

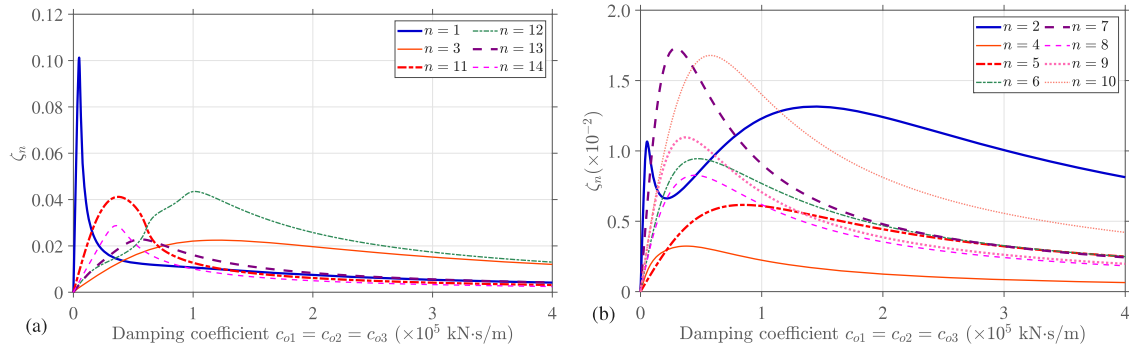


Fig. 12. Damping curves of the first 14 modes of the bridge with rigid outriggers and $c_{o1} = c_{o2} = c_{o3}$.

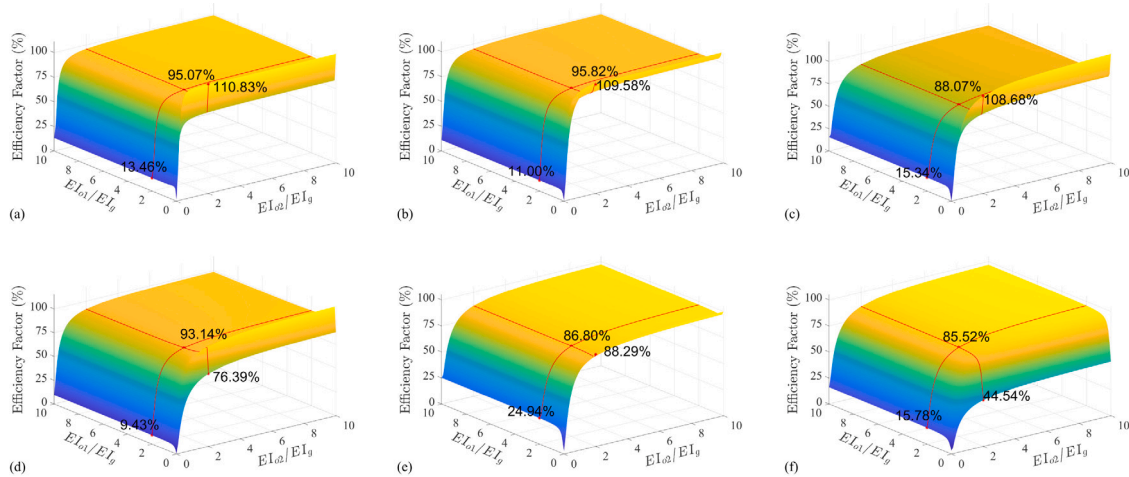


Fig. 13. Efficiency factor ρ of target modes with different stiffness ratio when DO1, DO2 and DO3 are installed: (a) 6th mode; (b) 8th mode; (c) 9th mode; (d) 10th mode; (e) 12th mode; (f) 13th mode.

Table 4
Optimized damper coefficients in Case 1 and Case 2.

DO number	Case 1 ($\times 10^5$ kN s/m)	Total c_o ($\times 10^5$ kN s/m)	Case 2 ($\times 10^5$ kN s/m)			Total c_o ($\times 10^5$ kN s/m)	Reduction rate (%)
	$c_{o1} = c_{o2} (= c_{o3})$		c_{o1}	c_{o2}	c_{o3}		
2	0.36	0.72	0.066	0.34	–	0.406	43.6
3	0.56	1.68	0.190	0.64	0.196	1.026	49.4

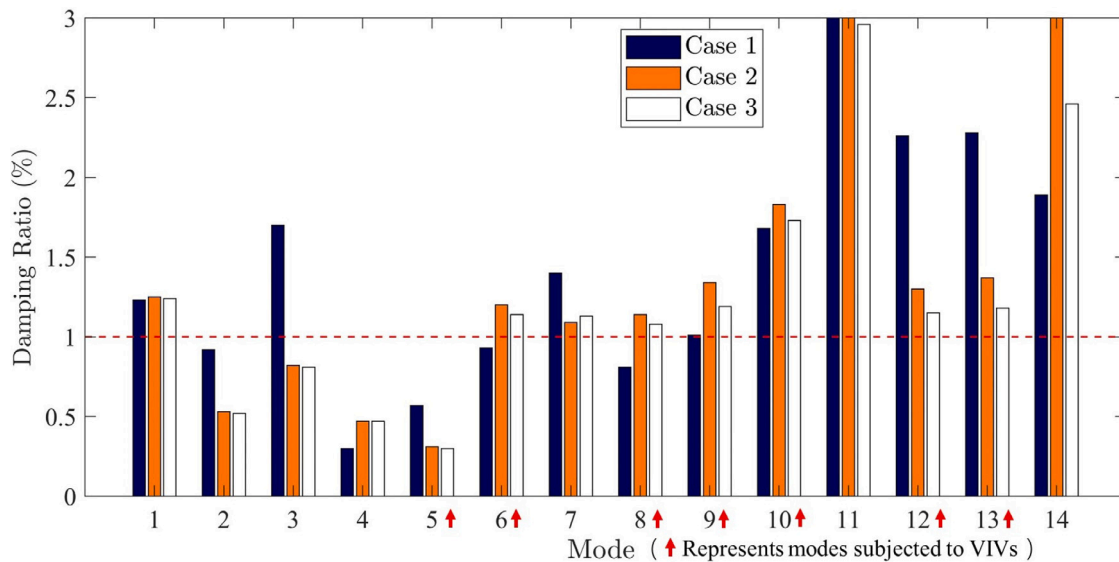


Fig. 14. A comparison of multimode damping effects realized by three damped outriggers with equal damper coefficients and with simultaneously optimized damper coefficients.

Table 5
Optimal damping ratio of the Xihoumen Bridge with three damped outriggers.

Mode no.	Frequency (Hz)	Damping ratio (%)			Efficiency factor ρ (%)
		Case 1	Case 2	Case3	
1	0.0945	1.23	1.25	1.24	99.3
2	0.1009	0.92	0.53	0.53	100.7
3	0.1315	1.70	0.82	0.81	98.2
4	0.1778	0.30	0.47	0.47	100.1
5 ^a	0.1838	0.57	0.31	0.30	99.6
6 ^a	0.2277	0.93	1.20	1.14	95.1
7	0.2570	1.40	1.09	1.13	102.8
8 ^a	0.2718	0.81	1.14	1.08	95.8
9 ^a	0.3249	1.01	1.34	1.19	88.1
10 ^a	0.3746	1.68	1.83	1.73	93.1
11	0.4062	3.38	3.12	2.96	98.6
12 ^a	0.4306	2.26	1.30	1.15	86.8
13 ^a	0.4825	2.28	1.37	1.18	85.5
14	0.5302	1.89	3.03	2.46	86.4

^arepresents the mode subjected to VIVs.

in practical design. Additionally, because sufficient damping and stiffness are provided to the main girders in the longitudinal direction by the three damped outriggers, longitudinal vibration components of the coupled modes vanish leading to purely vertical bending modes. It suggests that the applicability of replacing the original longitudinal dampers with damped outriggers.

5. Conclusions

The study presents an optimization design method to determine the optimal damper coefficients of damped outriggers installed at multiple positions of a bridge for improving multimode damping. The introduction of MAC for mode tracking and MOGA for solving the multi-objective multi-parameter optimization problem leads to a fully automated design procedure. The method is applied to the Xihoumen Bridge, a suspension bridge with a main span of 1650 m, for case study. The following conclusions are drawn.

- The optimization method can optimize damper coefficients of damped outriggers installed at different location. Particularly, the total damping coefficients decrease by 43.6% and 49.4% when damped outriggers are respectively installed at two and three

locations, as compared to the design adopting dampers of uniform size.

- For the suspension bridge with two damped outriggers, the optimization increases the minimal damping ratio of the target modes from 0.98% to 1.13% to meet the damping requirement.
- Outrigger stiffness is found influential on the damping effect in FE analysis. For the long-span suspension bridge, when the outrigger is 15 m high, the outrigger bending stiffness can be designed to be twice of the girder bending stiffness. The efficiency factors of the first 14 modes are above 85%, indicating the feasibility and effectiveness.
- The proposed optimization method is effective for the design of damped outriggers for bridge vibration mitigation. The method should be applicable to design other types of devices, such as MTMDs.

CRedit authorship contribution statement

Zhanhang Liu: Software, Methodology, Investigation, Writing – original draft. **Lin Chen:** Conceptualization, Software, Methodology, Investigation, Funding acquisition, Writing – original draft. **Limin Sun:** Conceptualization, Investigation, Supervision, Writing – review & editing. **Lin Zhao:** Resources, Methodology, Conceptualization, Writing – review & editing. **Wei Cui:** Resources, Methodology, Software, Writing – review & editing. **Hua Guan:** Resources, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] Fujino Y. Vibration, control and monitoring of long-span bridges—recent research, developments and practice in Japan. *J Construct Steel Res* 2002;58(1):71–97.
- [2] Dan D, Li H. Monitoring, intelligent perception, and early warning of vortex-induced vibration of suspension bridge. *Struct Control Health Monit* 2022;29(5):e2928.
- [3] Siringoringo DM, Fujino Y. Seismic response of a suspension bridge: Insights from long-term full-scale seismic monitoring system. *Struct Control Health Monit* 2018;25(11):e2252.
- [4] Gao W, Li G, Su Q, Han W. Impact of rigid central clamps on longitudinal deformation of long-span suspension bridges under vehicle excitations. *Struct Infrastruct Eng* 2022;18(6):760–74.
- [5] Li H, Laima S, Ou J, Zhao X, Zhou W, Yu Y, et al. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Eng Struct* 2011;33(6):1894–907.
- [6] Zhao L, Ge Y. Emergency measures for vortex-induced vibration of Humen bridge. In: *The 2020 world congress on advances in civil, environmental, & materials research*. ACEM20, Seoul, Korea; 2020, p. 25–8.
- [7] Hwang YC, Kim S, Kim H-K. Cause investigation of high-mode vortex-induced vibration in a long-span suspension bridge. *Struct Infrastruct Eng* 2020;16(1):84–93.
- [8] Ge Y, Xia J, Zhao L, Zhao S. Full aeroelastic model testing for examining wind-induced vibration of a 5,000 m spanned suspension bridge. *Front Built Environ* 2018;4(20):1–12.
- [9] Fujino Y, Kimura K, Tanaka H. Wind resistant design of bridges in Japan: developments and practices. Springer Japan; 2012.
- [10] Ge Y, Xu K, Zhao L. Review and reflection on vortex-induced vibration of main girders of long-span bridges. *China J Highw Transp* 2019;32(10):1–18 [in Chinese].
- [11] Li K, Qian G, Ge Y, Zhao L, Di J. Control effect and mechanism investigation on the horizontal flow-isolating plate for PI shaped bridge decks' VIV stability. *Wind Struct* 2019;28(2):99–110.
- [12] Zhao L, Li K, Wang C, Liu G, Liu T, Song S, et al. Review on passive aerodynamic countermeasures on main girders aiming at wind-induced stabilities of long-span bridges. *China J Highw Transp* 2019;32(10):34–48 [in Chinese].
- [13] Gao D, Deng Z, Yang W, Chen W. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: A vortex-dynamics approach. *J Fluids Struct* 2021;105:103348.
- [14] Ge Y, Zhao L, Cao J. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2022;220:104866.
- [15] Zhao L, Cui W, Shen X, Xu S, Ding Y, Ge Y. A fast on-site measure-analyze-suppress response to control vortex-induced-vibration of a long-span bridge. *Structures* 2022;35:192–201.
- [16] Fujino Y, Siringoringo D. Vibration mechanisms and controls of long-span bridges: a review. *Struct Eng Int* 2013;23(3):248–68.
- [17] Mokrani B, Tian Z, Alaluf D, Meng F, Preumont A. Passive damping of suspension bridges using multi-degree of freedom tuned mass dampers. *Eng Struct* 2017;153:749–56.
- [18] Ghaedi K, Ibrahim Z, Adeli H, Javanmardi A. Invited review: Recent developments in vibration control of building and bridge structures. *J Vibroeng* 2017;19(5):3564–80.
- [19] Xu K, Bi K, Han Q, Li X, Du X. Using tuned mass damper inerter to mitigate vortex-induced vibration of long-span bridges: Analytical study. *Eng Struct* 2019;182:101–11.
- [20] Xu K, Dai Q, Bi K, Fang G, Ge Y. Closed-form design formulas of TMDI for suppressing vortex-induced vibration of bridge structures. *Struct Control Health Monit* 2022;29(10):e3016.
- [21] Brzeski P, Lazarek M, Perlikowski P. Experimental study of the novel tuned mass damper with inerter which enables changes of inertance. *J Sound Vib* 2017;404:47–57.
- [22] Xu K, Dai Q, Bi K, Fang G, Zhao L. Multi-mode vortex-induced vibration control of long-span bridges by using distributed tuned mass damper inerters (DTMDIs). *J Wind Eng Ind Aerodyn* 2022;224:104970.
- [23] Zahrai SM, Froozanfar M. Performance of passive and active MTMDs in seismic response of Ahvaz cable-stayed bridge. *Smart Struct Syst* 2019;23(5):449–66.
- [24] Dai J, Xu Z, Xu W, Yan X, Chen Z. Multiple tuned mass damper control of vortex-induced vibration in bridges based on failure probability. *Structures* 2022;43:1807–18.
- [25] Hua X, Huang Z, Chen Z. Multi-mode vertical vortex-induced vibration of suspension bridges and control strategy. *China J Highw Transp* 2019;32(10):115–24 [in Chinese].
- [26] Chen Z, Hua X, Niu H, Huang Z, Zhang H, Chen J. Technological innovations in eddy current damping and its application in civil engineering. *China J Highw Transp* 2020;33(11):83–100 [in Chinese].
- [27] Preumont A, Voltan M, Sangiovanni A, Mokrani B, Alaluf D. Active tendon control of suspension bridges. *Smart Struct Syst* 2016;18(1):31–52.
- [28] Möller RN, Krenk S, Svendsen MN. Damping system for long-span suspension bridges. *Struct Control Health Monit* 2019;26(12):e2448.
- [29] Li H, Laima S, Zhang Q, Li N, Liu Z. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J Wind Eng Ind Aerodyn* 2014;124:54–67.
- [30] Ma C, Wang J, Li Q, Qin H, Liao H. Vortex-induced vibration performance and suppression mechanism for a long suspension bridge with wide twin-box girder. *J Struct Eng* 2018;144(11):04018202.
- [31] Dai J, Xu Z, Gai P, Hu Z. Optimal design of tuned mass damper inerter with a maxwell element for mitigating the vortex-induced vibration in bridges. *Mech Syst Signal Process* 2021;148:107180.
- [32] Bui H, Tran N. Multi-objective optimal design of TMDs for increasing critical flutter wind speed of bridges. *J Wind Eng Ind Aerodyn* 2022;225:104992.
- [33] Warburton GB. Optimum absorber parameters for various combinations of response and excitation parameters. *Earthq Eng Struct Dyn* 1982;10(3):381–401.
- [34] Chen Z, Huang Z, Wang J, Niu H. Basic requirements of tuned mass damper for bridges and the eddy current TMD. *J Hunan Univ: Nat Sci* 2013;40(8):6–10 [in Chinese].
- [35] Kwon S, Park K. Suppression of bridge flutter using tuned mass dampers based on robust performance design. *J Wind Eng Ind Aerodyn* 2004;92(11):919–34.
- [36] Han Q, Zhang X, Xu K, Du X. Free parameter optimization of DTMDs based on improved hybrid genetic-simulated annealing algorithm. *Int J Struct Stab Dyn* 2020;20(3):2050031.
- [37] Peng S, Zhang L, Liu Y, Li S. Parameters optimization and performance evaluation of multiple tuned mass dampers to mitigate the vortex-induced vibration of a long-span bridge. *Structures* 2022;38:1595–606.
- [38] Fadel Miguel LF, Piva dos Santos G. Optimization of multiple tuned mass dampers for road bridges taking into account bridge-vehicle interaction, random pavement roughness, and uncertainties. *Shock Vib* 2021;2021(6620427):1–17.
- [39] Miguel LF, Lopez RH, Torii AJ, Miguel LF, Beck AT. Robust design optimization of TMDs in vehicle-bridge coupled vibration problems. *Eng Struct* 2016;126:703–11.
- [40] Chen L, Liu Z, Nagarajaiah S, Sun L, Zhao L, Cui W. Vibration mitigation of long-span bridges with damped outriggers. *Eng Struct* 2022;271:114873.
- [41] Nagarajaiah S, Chen L, Wang M. Adaptive stiffness structures with dampers: seismic and wind response reduction using passive negative stiffness and inerter systems. *J Struct Eng* 2022;148(11):04022179.
- [42] Liu Z, Chen L, Sun L. Damping of long-span suspension bridges with damped outriggers. In: *IABSE congress, Nanjing 2022: bridges and structures: connections, integration and harmonisation*. Nanjing, China; 2022, p. 1856–64.
- [43] Zienkiewicz OC, Taylor RL, Zhu JZ. The finite element method: its basis and fundamentals. Elsevier; 2005.
- [44] Li M, Sun Y, Liao H, Meng F, Ma C. Vortex-induced vibration of steel box girder with large projecting slab and its mitigation countermeasures for Hong Kong-Zhuhai-Macao bridge. *J Tsinghua Univ, Sci. Technol.* 2020;60(1):57–65 [in Chinese].
- [45] Ansys Academic Research Mechanical, Release 19.2, Help System, Mechanical User's Guide. ANSYS, Inc.; 2018.
- [46] Vacher P, Jacquier B, Bucharles A. Extensions of the MAC criterion to complex modes. In: *Proceedings of ISMA2010 including USD2010*. 2010, p. 2713–25.
- [47] Siringoringo DM, Fujino Y. System identification of suspension bridge from ambient vibration response. *Eng Struct* 2008;30(2):462–77.
- [48] Kim TS, Kim YY. Mac-based mode-tracking in structural topology optimization. *Comput Struct* 2000;74(3):375–83.
- [49] Allemang RJ. The modal assurance criterion—twenty years of use and abuse. *J Sound Vib* 2003;37(8):14–23.
- [50] Yang X, He X. Nature-inspired optimization algorithms in engineering: overview and applications. In: *Nature-inspired computation in engineering*. Springer International Publishing; 2016, p. 1–20.
- [51] Kumar M, Husain M, Upreti N, Gupta D. Genetic algorithm: review and application. In: *International Journal of Information Technology and Knowledge Management*. 2, (2):World Scientific; 2010, p. 451–4.
- [52] Katoch S, Chauhan SS, Kumar V. A review on genetic algorithm: past, present, and future. *Multimedia Tools Appl* 2021;80(5):8091–126.
- [53] Di F, Chen L, Sun L. Optimal design of dampers for multi-mode cable vibration control based on genetic algorithm. *Int J Struct Stab Dyn* 2021;21(04):2150058.
- [54] Di F, Sun L, Chen L. Optimization of hybrid cable networks with dampers and cross-ties for vibration control via multi-objective genetic algorithm. *Mech Syst Signal Process* 2022;166:108454.
- [55] Zoltowski K, Banas A, Binczyk M, Kalitowski P. Control of the bridge span vibration with high coefficient passive damper. Theoretical consideration and application. *Eng Struct* 2022;254:113781.
- [56] Song H, Wang X. Zhoushan Xihoumen Bridge with the world record span length of steel box girder, China. *Struct Eng Int* 2010;20(3):312–6.
- [57] Ge Y. Wind resistance of long span suspension bridges. China Communications Press; 2011 [in Chinese].
- [58] Larsen A, Savage MG, Lafrenière A, Hui MC, Larsen SV. Investigation of vortex response of a twin box bridge section at high and low Reynolds numbers. *J Wind Eng Ind Aerodyn* 2008;96(6):934–44.
- [59] Li S, Laima S, Li H. Data-driven modeling of vortex-induced vibration of a long-span suspension bridge using decision tree learning and support vector regression. *J Wind Eng Ind Aerodyn* 2018;172:196–211.